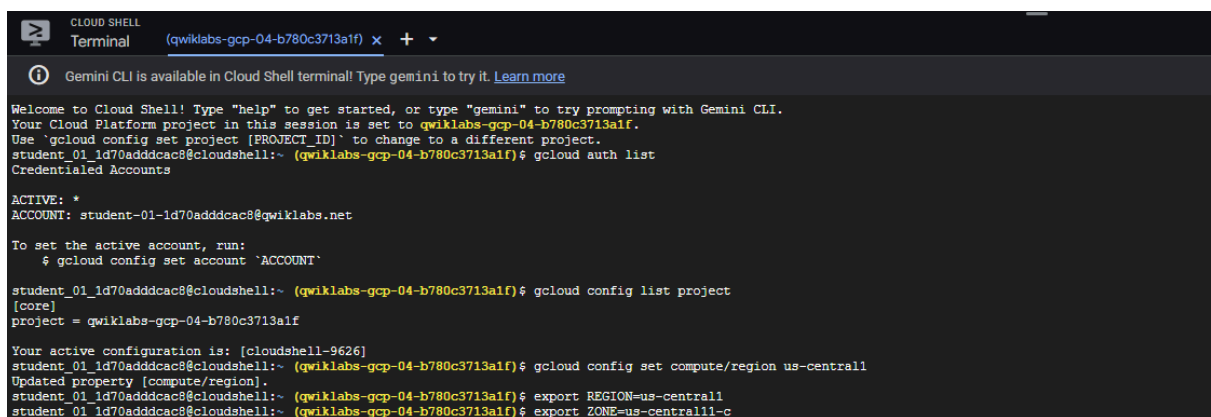
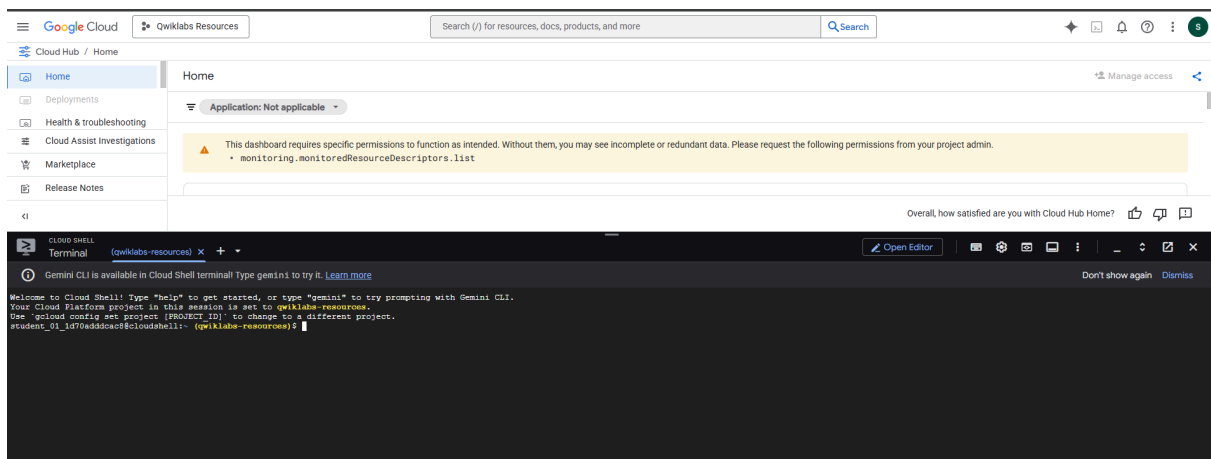


Create a VM on Google Cloud

Sur google Cloud nous pouvons creer plusieurs VM qui fonctionne avec different systeme d'exploitation (OS) nous retrouvons par exemple plusieurs distribution linux (Debian , Ubuntu , Suse , Red Hat , CoreOS ...)

Dans ce lab nous allons creer plusieurs instance de VM en utilisant la GoogleCloud console et les gcloud command line mais aussi connecter un NGINX web server a notre VM



← Create an instance Open project picker (Ctrl O) Create VM from...

- Machine configuration
e2-medium, us-central1-c
- OS and storage
Debian GNU/Linux 12 (bookworm)
- Data protection
Snapshot schedules
- Networking**
1 firewall rule, 1 network interface
- Observability
Install Ops Agent
- Security
- Advanced

Networking

Firewall ⓘ
Select firewall rules to allow specific network traffic from the Internet. Firewall rules will only be applied when an instance is created.

☒ Allow HTTP traffic
☐ Allow HTTPS traffic
☐ Allow Load Balancer Health Checks

Network tags
Network tags
http-server X ⓘ

Hostname
Hostname ⓘ
Set a custom hostname for this instance or leave it default. Choice is permanent

IP forwarding ⓘ
☐ Enable

Monthly estimate
\$25.46
That's about \$0.03 hourly
Pay for what you use: no upfront costs and per second billing

Item	Monthly estimate
2 vCPU + 4 GB memory	\$24.46
10 GB balanced persistent disk	\$1.00
Logging	Cost varies ⓘ
Monitoring	Cost varies ⓘ
Snapshot schedule	Cost varies ⓘ
Total	\$25.46

[Compute Engine pricing](#) ⓘ
[Cloud Operations pricing](#) ⓘ
[Less](#)

Google Cloud qwklabs-gcp-04-b780c3713a1f Search (J) for resources, docs, products, and more

Compute Engine VM instances Create Instance Import VM Refresh

Overview Security risk overview

Virtual machines VM instances Instance templates Sole-tenant nodes Machine images TPUs

VM instances

Filter Enter property name or value ⓘ ⓘ

Status	Name	Zone	Recommendations	In use by	Internal IP	External IP	Connect
☐	gcelab	us-central1-c			10.128.0.2 (nic0)	34.135.216.54 (nic0)	SSH

Related actions Hide

ssh.cloud.google.com/v2/ssh/projects/qwklabs-gcp-04-b780c3713a1f/zones/us-central1-c/instances/gcelab?authuser=0&hl=en_US&proj...

ssh.cloud.google.com/v2/ssh/projects/qwklabs-gcp-04-b780c3713a1f/zones/us-central1-c/instances/gcelab?authuser=0&hl=en_...

SSH-in-browser UPLOAD FILE DOWNLOAD FILE ⓘ ⌨ ⚙

```
The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Jan  7 15:10:59 2026 from 35.235.244.34
student-01-1d70adddcac@gcelab:~$ sudo apt-get update
Get:1 file:/etc/apt/mirrors/debian.list Mirrorlist [30 B]
Get:3 file:/etc/apt/mirrors/debian-security.list Mirrorlist [39 B]
Hit:7 https://packages.cloud.google.com/apt google-compute-engine-bookworm-stable InRelease
Get:8 https://packages.cloud.google.com/apt cloud-sdk-bookworm InRelease [1656 B]
Hit:2 https://deb.debian.org/debian bookworm InRelease
Get:4 https://deb.debian.org/debian bookworm-updates InRelease [55.4 kB]
Get:5 https://deb.debian.org/debian bookworm-backports InRelease [59.4 kB]
Get:6 https://deb.debian.org/debian-security bookworm-security InRelease [48.0 kB]
Get:9 https://packages.cloud.google.com/apt cloud-sdk-bookworm/main amd64 Packages [4462 kB]
Get:10 https://packages.cloud.google.com/apt cloud-sdk-bookworm/main all Packages [1931 kB]
Get:11 https://deb.debian.org/debian bookworm-backports/main Sources.diff/Index [63.3 kB]
Get:12 https://deb.debian.org/debian bookworm-backports/main amd64 Packages.diff/Index [63.3 kB]
Get:13 https://deb.debian.org/debian bookworm-backports/main Translation-en.diff/Index [63.3 kB]
Get:17 https://deb.debian.org/debian bookworm-backports/main Sources T-2025-12-27-0204.04-F-2025-12-12-2006.24.
pdiff [1885 B]
Get:18 https://deb.debian.org/debian bookworm-backports/main amd64 Packages T-2025-12-27-0204.04-F-2025-12-13-0
204.47.pdiff [3477 B]
Get:17 https://deb.debian.org/debian bookworm-backports/main Sources T-2025-12-27-0204.04-F-2025-12-12-2006.24.
pdiff [1885 B]
Get:18 https://deb.debian.org/debian bookworm-backports/main amd64 Packages T-2025-12-27-0204.04-F-2025-12-13-0
204.47.pdiff [3477 B]
Get:19 https://deb.debian.org/debian bookworm-backports/main Translation-en T-2025-12-13-0204.47-F-2025-12-13-0
204.47.pdiff [1759 B]
Get:19 https://deb.debian.org/debian bookworm-backports/main Translation-en T-2025-12-13-0204.47-F-2025-12-13-0
204.47.pdiff [1759 B]
Get:14 https://deb.debian.org/debian-security bookworm-security/main Sources [196 kB]
Get:15 https://deb.debian.org/debian-security bookworm-security/main amd64 Packages [290 kB]
Get:16 https://deb.debian.org/debian-security bookworm-security/main Translation-en [176 kB]
Fetched 7415 kB in 1s (6052 kB/s)
Reading package lists... Done
student-01-1d70adddcac@gcelab:~$
```

Connect SSH

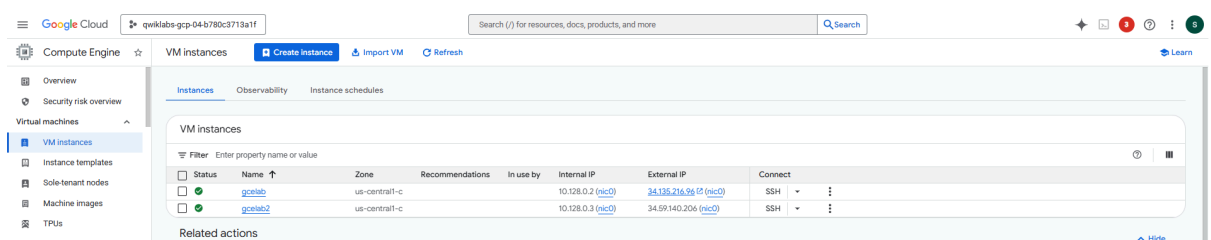
```
Welcome to Cloud Shell! Type "help" to get started, or type "gcp" to try prompting with Gemini CLI.
Your cloud platform project in this session is set to qwiklabs-gcp-04-b780c3713a1f.
Use "gcloud config set project [PROJECT_ID]" to change to a different project.
student_01_1d70edddac@cloudshell:~ (qwiklabs-gcp-04-b780c3713a1f) $ gcloud auth list
Credentialed Accounts

ACTIVE: *
ACCOUNT: student-01-1d70edddac@qwiklabs.net

To set the active account, run:
$ gcloud config set account 'ACCOUNT'

student_01_1d70edddac@cloudshell:~ (qwiklabs-gcp-04-b780c3713a1f) $ gcloud config list project
[core]
project = qwiklabs-gcp-04-b780c3713a1f

Your active configuration is: [cloudshell-14624]
student_01_1d70edddac@cloudshell:~ (qwiklabs-gcp-04-b780c3713a1f) $ gcloud config set compute/region us-central1
Updated property [compute/region].
student_01_1d70edddac@cloudshell:~ (qwiklabs-gcp-04-b780c3713a1f) $ export REGION=us-central1
student_01_1d70edddac@cloudshell:~ (qwiklabs-gcp-04-b780c3713a1f) $ export ZONE=us-central1-c
student_01_1d70edddac@cloudshell:~ (qwiklabs-gcp-04-b780c3713a1f) $ gcloud compute instances create gcp-lab2 --machine-type e2-medium --zone=$ZONE
Created [https://www.googleapis.com/compute/v1/projects/qwiklabs-gcp-04-b780c3713a1f/zones/us-central1-c/instances/gcp-lab2].
NAME: gcp-lab2
ZONE: us-central1-c
MACHINE_TYPE: e2-medium
PREEMPTIBLE:
INTERNAL_IP: 10.128.0.3
EXTERNAL_IP: 34.59.140.206
STATUS: RUNNING
student_01_1d70edddac@cloudshell:~ (qwiklabs-gcp-04-b780c3713a1f) $ ||
```



GSP001



Overview

Compute Engine allows you to create virtual machines (VMs) that run different operating systems, including multiple flavors of Linux (Debian, Ubuntu, Suse, Red Hat, CoreOS) and Windows Server, on Google infrastructure. You can run thousands of virtual CPUs on a system that is designed to be fast and to offer strong consistency of performance.

In this hands-on lab, you create VM instances of various machine types using the Google Cloud console and the `gcloud` command line in Cloud Shell. You also learn how to connect an NGINX web server to your VM.

Although you can easily copy and paste commands from the lab to the appropriate place, it is recommended that you type the commands yourself to reinforce your understanding of the core concepts.

Objectives

In this lab, you learn how to perform the following tasks:

- Create a VM with the Cloud console.

- Create a VM with the `gcloud` command line.
- Deploy a web server and connect it to a VM.

Prerequisites

- Familiarity with standard Linux text editors such as `vim`, `emacs`, or `nano`.

Setup and requirements

Before you click the Start Lab button

Read these instructions. Labs are timed and you cannot pause them. The timer, which starts when you click **Start Lab**, shows how long Google Cloud resources are made available to you.

This hands-on lab lets you do the lab activities in a real cloud environment, not in a simulation or demo environment. It does so by giving you new, temporary credentials you use to sign in and access Google Cloud for the duration of the lab.

To complete this lab, you need:

- Access to a standard internet browser (Chrome browser recommended).

Note: Use an Incognito (recommended) or private browser window to run this lab. This prevents conflicts between your personal account and the student account, which may cause extra charges incurred to your personal account.

- Time to complete the lab—remember, once you start, you cannot pause a lab.

Note: Use only the student account for this lab. If you use a different Google Cloud account, you may incur charges to that account.

How to start your lab and sign in to the Google Cloud console

1. Click the **Start Lab** button. If you need to pay for the lab, a dialog opens for you to select your payment method. On the left is the Lab Details pane with the following:

- The Open Google Cloud console button
 - Time remaining
 - The temporary credentials that you must use for this lab
 - Other information, if needed, to step through this lab
2. Click **Open Google Cloud console** (or right-click and select **Open Link in Incognito Window** if you are running the Chrome browser).

The lab spins up resources, and then opens another tab that shows the Sign in page.

Tip: Arrange the tabs in separate windows, side-by-side.

Note: If you see the **Choose an account** dialog, click **Use Another Account**.

3. If necessary, copy the **Username** below and paste it into the **Sign in** dialog.Copied!

student-01-1d70adddcac8@qwiklabs.net

You can also find the Username in the Lab Details pane.

4. Click **Next**.
5. Copy the **Password** below and paste it into the **Welcome** dialog.Copied!

hSPj1qQlCx0c

You can also find the Password in the Lab Details pane.

6. Click **Next**.

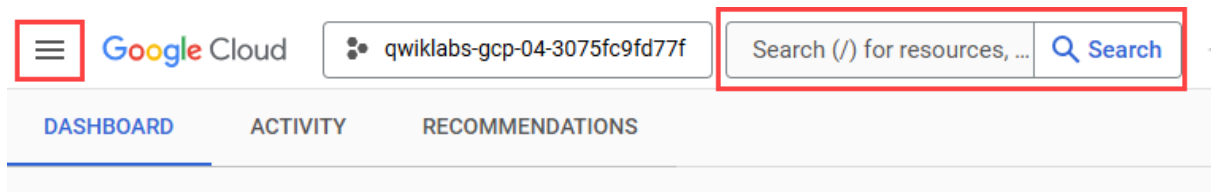
Important: You must use the credentials the lab provides you. Do not use your Google Cloud account credentials.

Note: Using your own Google Cloud account for this lab may incur extra charges.

7. Click through the subsequent pages:
 - Accept the terms and conditions.
 - Do not add recovery options or two-factor authentication (because this is a temporary account).
 - Do not sign up for free trials.

After a few moments, the Google Cloud console opens in this tab.

Note: To access Google Cloud products and services, click the **Navigation menu** or type the service or product name in the **Search** field.



Activate Cloud Shell

Cloud Shell is a virtual machine that is loaded with development tools. It offers a persistent 5GB home directory and runs on the Google Cloud. Cloud Shell provides command-line access to your Google Cloud resources.

1. Click **Activate Cloud Shell** at the top of the Google Cloud console.



2. Click through the following windows:

- Continue through the Cloud Shell information window.
- Authorize Cloud Shell to use your credentials to make Google Cloud API calls.

When you are connected, you are already authenticated, and the project is set to your **Project_ID**, `qwiklabs-gcp-04-b780c3713a1f`. The output contains a line that declares the **Project_ID** for this session:

```
Your Cloud Platform project in this session is set to qwiklabs-gcp-04-b780c3713a1f
```

`gcloud` is the command-line tool for Google Cloud. It comes pre-installed on Cloud Shell and supports tab-completion.

1. (Optional) You can list the active account name with this command:

```
gcloud auth list
```

Copied!

1. Click **Authorize**.

Output:

```
ACTIVE: *  
ACCOUNT: student-01-1d70adddcac8@qwiklabs.net
```

To set the active account, run:

```
$ gcloud config set account 'ACCOUNT'
```

1. (Optional) You can list the project ID with this command:

```
gcloud config list project
```

Copied!

Output:

```
[core]
```

```
project = qwiklabs-gcp-04-b780c3713a1f
```

Note: For full documentation of `gcloud`, in Google Cloud, refer to [the gcloud CLI overview guide](#).

Understanding Regions and Zones

Certain Compute Engine resources live in regions or zones. A region is a specific geographical location where you can run your resources. Each region has one or more zones. For example, the `us-central1` region denotes a region in the Central United States that has zones `us-central1-a`, `us-central1-b`, `us-central1-c`, and `us-central1-f`.

Regions	Zones
Western US	us-west1-a, us-west1-b
Central US	us-central1-a, us-central1-b, us-central1-d, us-central1-f
Eastern US	us-east1-b, us-east1-c, us-east1-d
Western Europe	eu-west1-b, eu-west1-c, eu-west1-d
Eastern Asia	asia-east1-a, asia-east1-b, asia-east1-c

Resources that live in a zone are referred to as zonal resources. Virtual machine Instances and persistent disks live in a zone. To attach a persistent disk to a virtual machine instance, both resources must be in the same zone. Similarly, if you want to assign a static IP address to an instance, the instance must be in the same region as the static IP.

Learn more about regions and zones and see a complete list in the Compute Engine page, [Regions and zones documentation](#).

Set the region and zone

1. Run the following command to set the project region for this lab:


```
gcloud config set compute/region us-central1
```

Copied!

1. To create a variable for region, run the following command:

```
export REGION=us-central1
```

Copied!

1. Run the following command to create a variable for the zone:

```
export ZONE=us-central1-c
```

Copied!

Learn more from the [Regions and zones](#) documentation.

Note: When you run `gcloud` on your own machine, the config settings are persisted across sessions. But in Cloud Shell, you need to set this for every new session or reconnection.

Task 1. Create a new instance from the Cloud console

In this section, you create new predefined machine types with Compute Engine from the Cloud console.

1. In the **Cloud console**, on the **Navigation menu** ($\equiv$), click **Compute Engine > VM Instances**.

┆ This may take a minute to initialize for the first time.

2. To create a new instance, click **Create Instance**.
3. In the **Machine configuration**:

Enter the values for the following fields:

Field	Value	Additional Information
Name	<code>gcelab</code>	Name for the VM instance
Region	<code>us-central1</code>	For more information about regions, see the Compute Engine guide, Regions and Zones .
Zone	<code>us-central1-c</code>	Note: Remember the zone that you selected to use later. For more information about zones, see the Compute Engine guide, Regions and Zones .

Field	Value	Additional Information
Series	E2	
Machine Type	e2-medium	This is an e2-medium, 2-CPU, 4GB RAM instance. Several machine types are available, ranging from micro instance types to 32-core/208GB RAM instance types. For more information, see the Compute Engine guide, About machine families .

Note: A new project has a default resource quota, which may limit the number of CPU cores. You can request more when you work on projects outside this lab.

1. Click **OS and storage**.

Click **Change** to begin configuring your boot disk and select the values for:

- **Operating system:** Debian
- **Version:** Debian GNU/Linux 12 (bookworm)
- **Boot disk type:** Balanced persistent disk
- **Size:** 10 GB

Several images are available, including Debian, Ubuntu, CoreOS, and premium images such as Red Hat Enterprise Linux and Windows Server. For more information, see the [Operating System documentation](#).

2. Click **Networking**.

- **Firewall:** Allow HTTP traffic

Select this option in order to access a web server that you install later.

Note: This automatically creates a firewall rule to allow HTTP traffic on port 80.

3. Once all sections are configured, scroll down and click **Create** to launch your new virtual machine instance.

It should take about a minute for the VM, `gcelab`, to be created.

After `gcelab` is created, the **VM Instances** page lists it in the VM instances list.

4. To use SSH to connect to the VM, click **SSH** to the right of the instance name, `gcelab`.

This launches an SSH client directly from your browser.

Note: Learn more about how to use SSH to connect to an instance from the Compute Engine guide, [Connect to Linux VMs using Google tools](#).

Task 2. Install an NGINX web server

Now you install an NGINX web server, one of the most popular web servers in the world, to connect your VM to something.

1. Run the following command to update the OS: Copied!

```
sudo apt-get update
```

Expected output:

```
Get:1 file:/etc/apt/mirrors/debian.list Mirrorlist [30 B]
Get:5 file:/etc/apt/mirrors/debian-security.list Mirrorlist [39 B]
Get:7 https://packages.cloud.google.com/apt google-compute-engine-bookworm-stable InRelease [1321 B]
Get:2 https://deb.debian.org/debian bookworm InRelease [151 kB]
Get:3 https://deb.debian.org/debian bookworm-updates InRelease [55.4 kB]
Get:4 https://deb.debian.org/debian bookworm-backports InRelease [59.0 kB]
Hit:8 https://packages.cloud.google.com/apt cloud-sdk-bookworm InRelease
Hit:6 https://deb.debian.org/debian-security bookworm-security InRelease
Fetched 267 kB in 1s (274 kB/s)
Reading package lists... Done
```

2. Run the following command to install NGINX: Copied!

```
sudo apt-get install -y nginx
```

Expected output:

```
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following NEW packages will be installed:
...
```

3. Run the following command to confirm that NGINX is running: Copied!

```
ps auxx | grep nginx
```

Expected output:

```
root    2330  0.0  0.0 159532 1628 ?        Ss   14:06   0:00 nginx: master process /usr/sbin/nginx -g daemon
on; master_process on;
www-data 2331  0.0  0.0 159864 3204 ?        S    14:06   0:00 nginx: worker process
www-data 2332  0.0  0.0 159864 3204 ?        S    14:06   0:00 nginx: worker process
root    2342  0.0  0.0 12780   988 pts/0    S+   14:07   0:00 grep nginx
```

4. To see the web page, return to the Cloud console and click the **External IP** link in the row for your machine, or add the **External IP** value to `http://EXTERNAL_IP/` in a new browser window or tab. *Assessment Completed!*

A default web page should open that says: **Welcome to nginx!**

To check your progress in this lab, click **Check my progress** below. A checkmark means you're successful.

Assessment Completed!

Create a Compute Engine instance and add an NGINX Server to your instance with necessary firewall rules.

Task 3. Create a new instance with gcloud

Instead of using the Cloud console to create a VM instance, use the command line tool `gcloud`, which is pre-installed in [Google Cloud Shell](#). Cloud Shell is an interactive shell environment for Google Cloud loaded with all the development tools you need (`gcloud`, `git`, and others) and offers a persistent 5-GB home directory.

Note: If you want to try this on your own machine, read the [gcloud command line tool guide](#).

1. In Cloud Shell, run the following `gcloud` command to create a new VM instance from the command line: *Copied!Assessment Completed!*

```
gcloud compute instances create gcelab2 --machine-type e2-medium --zone=$ZONE
```

Expected output:

```
Created [...gcelab2].
NAME: gcelab2
ZONE: us-central1-c
MACHINE_TYPE: e2-medium
PREEMPTIBLE:
INTERNAL_IP: 10.128.0.3
EXTERNAL_IP: 34.136.51.150
STATUS: RUNNING
```

To check your progress in this lab, click **Check my progress** below. A checkmark means you're successful.

Assessment Completed!

Create a new instance with gcloud.

The new instance has these default values:

- The [Debian GNU/Linux 12 \(bookworm\)](#) image.
- The `e2-medium` machine type.

- A root persistent disk with the same name as the instance; the disk is automatically attached to the instance.

When working in your own project, you can specify a custom machine type.

2. To see all the defaults, run the following command: Copied!

```
gcloud compute instances create --help
```

Note: You can set the default region and zones that `gcloud` uses if you are always working within one region/zone and you don't want to append the `--zone` flag every time.

To do this, run these commands:

```
gcloud config set compute/zone ...
gcloud config set compute/region ...
```

3. To exit `help`, press **CTRL+C**.
4. In the Cloud console, on the **Navigation menu** (), click **Compute Engine > VM instances**. Or if you still had the VM instances page open, just click **Refresh**. Your two new instances should be listed.



5. You can also use SSH to connect to your instance via `gcloud`. Make sure to add your zone, or omit the `-zone` flag if you've set the option globally: Copied!

```
gcloud compute ssh gcelab2 --zone=us-central1-c
```

6. Enter **Y** to continue.

```
Do you want to continue? (Y/n)
```

7. Press **Enter** through the passphrase section to leave the passphrase empty.

```
Generating public/private rsa key pair.
Enter passphrase (empty for no passphrase)
```

8. After connecting, disconnect from SSH by exiting from the remote shell with the command that follows: Copied!

```
exit
```

Task 4. Test your knowledge

Test your knowledge about Google Cloud by taking the quiz. (Please select multiple correct options if necessary.)

Through which of the following ways can you create a VM instance in Compute Engine?checkThe gcloud command line toolcheckThe Cloud console

Congratulations!

Compute Engine is the foundation of Google Cloud's infrastructure as a service. You created a VM with Compute Engine and can now map your existing server infrastructure, load balancers, and network topology to Google Cloud.

Take your next lab

Continue your course with [Get Started with Cloud Shell and gcloud](#), or check out the following lab:

- [Provision Services with Google Cloud Marketplace](#)

Next steps / Learn more

- For an overview of VMs, refer to the [Virtual machine instances](#) documentation.
- Check out how to [migrate VMs to the Google Cloud](#).
- Learn more about [VPC networks](#).

Google Cloud training and certification

...helps you make the most of Google Cloud technologies. [Our classes](#) include technical skills and best practices to help you get up to speed quickly and continue your learning journey. We offer fundamental to advanced level training, with on-demand, live, and virtual options to suit your busy schedule. [Certifications](#) help you validate and prove your skill and expertise in Google Cloud technologies.

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